

POF Calculus Primer

With Worked Example: ± 0 Collapse and Entanglement

Prepared for: Joe

Project: Neural Optical Co-Processor

POF Calculus Recap

Recap: POF Calculus Operators

- Probability Operator (integral form): accumulates amplitudes across origins.
- Higgs Operator (derivative form): localizes system to a definite origin (vev).
- Fundamental Theorem of POF (FTPOF): Higgs collapse (derivative) is dual to probabil

Worked Example: ± 0 Collapse

Worked Example: ± 0 Superposition Collapse

****Setup:****

System state: $|\psi\rangle = (|+0\rangle + |-0\rangle)/\sqrt{2}$

Origin states: $|0+\rangle, |0-\rangle$ (two possible origin branches)

****Step 1: Joint state in POF:****

$$|\Psi\rangle = (|+0\rangle \otimes |0+\rangle + |-0\rangle \otimes |0-\rangle)/\sqrt{2}$$

****Step 2: Probability Operator (integral form):****

$$P(+0,0+) = 1/2, \quad P(-0,0-) = 1/2$$

This marginalizes over both branches $\rightarrow$ equal distribution.

****Step 3: Higgs Operator (derivative form):****

Collapse = alignment with one origin branch.

If origin aligns to $|0+\rangle \rightarrow$ system collapses to $|+0\rangle$.

If origin aligns to $|0-\rangle \rightarrow$ system collapses to $|-0\rangle$.

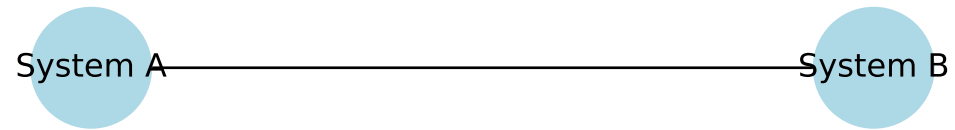
****Step 4: FTPOF consistency check:****

- Probability operator gave integral distribution (50/50 across branches).

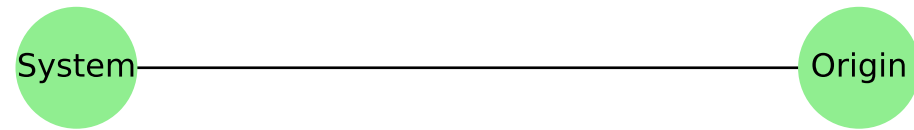
- Higgs operator localized one outcome (derivative selection).

Together they form a dual description of collapse.

Standard Bell Entanglement
 $(|00\rangle + |11\rangle)/\sqrt{2}$



POF Entanglement
 $(|+0,O+\rangle + |-0,O-\rangle)/\sqrt{2}$



Entanglement and FTPOF

Entanglement and the Fundamental Theorem of POF

Standard Bell entanglement:

$$|\Psi\rangle = (|0\rangle|0\rangle + |1\rangle|1\rangle)/\sqrt{2}$$

shows correlations between two systems.

POF entanglement:

$$|\Psi\rangle = (|+0\rangle\otimes|0+\rangle + |-0\rangle\otimes|0-\rangle)/\sqrt{2}$$

shows correlations between a system and its origin.

Relation to FTPOF:

- Probability operator (integral): spreads amplitude across both branches.
- Higgs operator (derivative): selects one branch (collapse).

Thus entanglement is the natural arena of the Fundamental Theorem of POF: distribution

Implication:

- Some entanglement is spurious (from shared origin uncertainty).
- Genuine entanglement remains after origin alignment.

In POF, entanglement is not mysterious nonlocality but the expression of system–origin